

LINES OF CODES(LOC):

| File Name               | Estimated LOC | Primary Responsibility                        |
|-------------------------|---------------|-----------------------------------------------|
| AdminController.php     | 372           | Managing users, tools, and system logs.       |
| AuthController.php      | 95            | User authentication, login, and registration. |
| FarmerController.php    | 703           | Handling plot sharing and mentorship logic.   |
| VolunteerController.php | 166           | Coordinating volunteer shifts and hours.      |
| DBController.php        | 60            | Core database connection and CRUD operations. |
| RentController.php      | 25            | Processing plot availability and rentals.     |

OO COMPLEXITY:

| Function Name | Parent Controller   | CCM | WMC | Risk Level   |
|---------------|---------------------|-----|-----|--------------|
| login()       | AuthController      | 6   | 6   | Low (Simple) |
| addTool()     | AdminController     | 5   | 5   | Low (Simple) |
| joinShift()   | VolunteerController | 6   | 6   | Low (Simple) |
| share_plot()  | FarmerController    | 8   | 8   | Moderate     |

## Unit Testing :

### Authentication (login):

| Test Case ID | Scenario         | Input (Mocked DB Result)             | Expected Outcome                                               |
|--------------|------------------|--------------------------------------|----------------------------------------------------------------|
| AUTH-01      | Successful Login | Returns 1 row with user data.        | <code>\$_SESSION[ 'userId' ]</code> is set; returns true.      |
| AUTH-02      | Wrong Password   | Returns 0 rows (Empty array).        | <code>\$_SESSION[ 'errMsg' ]</code> is set; returns false.     |
| AUTH-03      | Database Error   | <code>select()</code> returns false. | Function returns false early.                                  |
| AUTH-04      | SQL Injection    | Input contains ' OR 1=1.             | Verify query string is escaped (if using prepared statements). |

Authorization (Role Check) :

| Test Case ID   | Scenario           | User Role       | Expected Access                                  |
|----------------|--------------------|-----------------|--------------------------------------------------|
| <b>AUTO-01</b> | Admin Access       | role = 'admin'  | Allowed to call AdminController methods.         |
| <b>AUTO-02</b> | Client Restriction | role = 'client' | Blocked from AdminController (Redirect to Home). |
| <b>AUTO-03</b> | Guest Access       | session = null  | Redirect to login.php.                           |